

SAMARA PIRES

Machine Learning — Applied AI — Data Science — Data Analytics

Edmonton, AB | samarapires.da@gmail.com | +1 (587) 982-1090

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Google Scholar](#)

EDUCATION

University of Alberta

Master of Science in Modelling, Data & Predictions; GPA: 3.7/4.0

Edmonton, AB

2025 – Jan 2027

University of Mumbai – Don Bosco Institute of Technology

Bachelor of Engineering in Computer Science; Runner-Up, Best Final Year Thesis

Mumbai, India

2017 – 2021

EXPERIENCE

AltaML

Associate Business Solutions Consultant

Edmonton, AB

May 2026 – Aug 2026

- Bridged business and technical teams in a Product Management-AI Architecture capacity, translating stakeholder needs into AI solution requirements, workflows, and product recommendations.
- Contributed to an LLM-powered Research Assistant for the Government of Alberta by gathering requirements, evaluating solution performance, and translating client feedback into product improvements.
- Collaborated with AI engineers to design, test, and refine agentic AI workflows using LLMs, retrieval-based approaches, prompt engineering, and system evaluation.

Amii – Alberta Machine Intelligence Institute

AICAP Level Up WILO Participant

Edmonton, AB

Sep 2025 – Present

- Selected for Amii's Level Up Work-Integrated Learning Opportunity, working alongside an Amii Machine Learning Scientist to analyze real-world AI challenges faced by startups.
- Conducted technical research across LLM agents, human-in-the-loop systems, prompt/context engineering, and agentic workflows, translating findings into technical recommendations.

Data for Good Edmonton

Data Analyst

Edmonton, AB

Sep 2025 – May 2026

- Built a Python/NumPy cost-benefit modelling pipeline to estimate the 10-year health and economic impact of Edmonton's protected cycling infrastructure.
- Conducted multi-scenario CVD and diabetes cost-savings analysis and developed visualizations comparing projected healthcare benefits with infrastructure costs.

LTIMindtree – Client: Johnson Controls

Quality Analyst

Mumbai, India

Jun 2021 – Oct 2024

- Developed automated testing frameworks using Python, Selenium, and BrowserStack, reducing manual testing effort by **81%**.
- Led cross-functional delivery across QA, development, and business teams, achieving a **5/5 client rating** and **100% client satisfaction**.

Expensing

Data Analyst Intern

Mumbai, India

Dec 2019 – Jan 2020

- Built Power BI dashboards and analyzed expense data to generate financial insights contributing to **23% cost savings**; automated data workflows to improve process efficiency by **32%**.

PROJECTS

MonteCore Finance – AI-Powered Financial Risk Intelligence Platform [Link](#) Python, Scikit-learn, Agentic AI, IBM watsonx.ai, Streamlit

- Built an end-to-end financial risk intelligence platform combining supervised ML, anomaly detection, and agentic AI across AML detection, transaction anomalies, and financial early-warning risk.
- Developed a grounded multi-agent investigation workflow using Mistral and Llama 3.3, with independent second-look review and human-in-the-loop decision support to reduce unsupported AI conclusions.

Budget Bites – RAG-Powered AI Recipe Planner [Link](#)

Python, LLMs, RAG, FAISS, Streamlit

- Built a Retrieval-Augmented Generation system using LLMs, FAISS, semantic search, and prompt engineering to retrieve relevant context and generate personalized recipe recommendations.

BNPL Risk Prediction – Bayesian Machine Learning [Link](#)

Python, PyMC, Bayesian Statistics

- Developed a Bayesian predictive model for Buy Now, Pay Later risk assessment and evaluated predictive uncertainty to generate interpretable customer risk estimates.

2026 Data Masters Challenge – 2nd Place [Link](#)

Python, Pandas, NumPy, Scikit-learn

- Built an end-to-end ML pipeline covering data cleaning, EDA, feature preparation, predictive modeling, evaluation, and visualization, contributing to a **2nd-place finish**.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, Pandas, NumPy, Data Cleaning, Data Transformation

Machine Learning & Statistics: Scikit-learn, PyMC, Predictive Modeling, Statistical Modeling, Bayesian Modeling, Model Evaluation

Applied AI: Large Language Models, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Semantic Search

Analytics & Visualization: Excel, Power BI, Tableau, Matplotlib, Exploratory Data Analysis

Tools & Platforms: Git, GitHub, Jupyter, Streamlit, Google BigQuery

Certifications: [Google Generative AI Leader](#), [Google AI Agents](#), [Google BigQuery Data Warehouse](#), [UC Davis SQL for Data Science](#)

RESEARCH & PUBLICATIONS

Cyber Threat Analysis Using Pearson & Spearman Correlation via EDA [Publication](#)

ICSCCC 2023

- Applied exploratory data analysis and Pearson and Spearman correlation techniques to analyze relationships within cybersecurity data and support statistical cyber-threat analysis.

Real-Time Position Monitoring System for Fall Detection & Analysis [Publication](#)

IEEE ICNTE 2021

- Developed research focused on real-time position monitoring and data-driven fall detection, applying machine learning techniques to identify and analyze fall-related events.

Fall Stop: Real-Time Fall Detection & Vital Monitoring [Publication](#)

IEEE INDISCON 2021

- Developed a real-time fall detection and vital-monitoring research system integrating machine learning, computer vision, and IoT-based monitoring for automated fall detection.